SHIVAM LALAKIYA

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Professional Summary

- Data Scientist with **4+ years** of experience building analytics that reach production and stay there: propensity and forecasting models deployed into live CRM operations, unattended data quality and reporting pipelines serving multiple teams, and LLM applications on secure enterprise infrastructure.

Experience

Washington University in St. Louis **Feb 2024 – Present**

Data Analyst, Grateful Patient Program *St. Louis, MO*

- Built an internal **AI analytics assistant** (**Next.js**, **LangGraph**, **MCP**, vector retrieval) on the university's secure AI gateway, indexing live **Salesforce** data and internal GitHub repositories to answer analytical questions in seconds; shipped from a standing start in **10 weeks**.
- Developed and productionized donor **propensity models** achieving a **66x lift** at the top of the ranked list, deployed into the CRM as interest codes now driving live gift-officer targeting alongside scoring and tiering frameworks adopted across Prospect Management and Research.
- Architected an automated **data quality pipeline** (**SQL Server Agent**, **16** checks across **11** CRM objects) serving **4** teams at **$0** infrastructure cost, with **519** runs over **17** failure-free weeks cutting missing contact preferences **73%** and unlinked accounts **79%**.
- Engineered **fuzzy matching** and duplicate-detection engines scanning a **703K-row** contact table weekly, surfacing **2,100+** probable duplicate pairs per cycle and unifying external VSE records with the CRM.
- Architected a multi-model cash-flow forecasting suite (**ARIMA, LSTM, Prophet**) over **26** fiscal years of actuals with macroeconomic inputs and multi-tier recession stress-testing, delivering **5-year** projections to Treasury and the **Board of Trustees**.
- Automated **8** recurring production reporting systems and **100+** custom reports across **50** departments, collapsing a **4-hour** manual cycle to **under 5 minutes** with zero manual rebuilds.

Invicro, LLC. **Sept 2023 – Feb 2024**

Associate Data Manager *Boston, MA*

- Engineered an automated data pipeline with **Apache Airflow** for pathological image transfer, reducing processing time by **20%** and ensuring on-time delivery for clinical research.
- Deployed **computer vision** models for pathological image analysis, achieving a **10%** improvement in bio-marker identification and contributing to a **15%** increase in study sponsors.
- Architected data transformation models in **dbt**, streamlining workflows and driving a **25% reduction** in downstream reporting errors.

Ring Therapeutics **May 2022 – Dec 2022**

Data Science Co-op *Boston, MA*

- Leveraged **NLP** (positional encoding) and **Graph Neural Networks** to predict gene tropism at a **0.75 F1-score**, generating vectors for targeted cell therapy delivery.
- Fine-tuned a **Protein-BERT** model on proprietary Anello sequences, reaching a **0.8 F1-score** for tropism classification, and automated **90%** of data loading via **Dockerized Airflow** DAGs.

Technical Skills

Languages: Python, SQL (T-SQL), R, Shell Scripting, Java, MATLAB
ML & AI: Forecasting (ARIMA, LSTM, Prophet), Propensity Modeling, NLP & Transformers (BERT), LLM Applications (RAG, LangGraph, MCP, Agents), Computer Vision
Data Engineering: ETL, Apache Airflow, dbt, SQL Server Agent, Docker, AWS, Kafka, REST APIs, Git, CI/CD
Platforms & BI: Salesforce, SQL Server, Tableau, Power BI, React/Next.js

Education

Northeastern University, Boston, MA **Sept 2021 – July 2023**

Master of Science in Data Analytics Engineering *GPA: 3.8/4.0*

Sardar Vallabhbhai National Institute of Technology (NIT Surat), India **July 2017 – May 2021**

Bachelor of Technology in Electronics and Communication Engineering *GPA: 8.1/10*

Open Source

- **PhilanthroPy** (author & maintainer): **scikit-learn**-style donor analytics library for nonprofit fundraising (donor scoring, lifecycle segmentation, campaign metrics). **v0.8.0**, **265** PyPI downloads/mo, **19** contributors, **2,671** tests.
- **26 merged pull requests** to **LlamaIndex** (52K stars), **sktime** (10K), **MAPIE**, **PyMC-Marketing**, and **TorchCP**, spanning memory backends, vector-store metadata, and conformal prediction.